



quantitative chemistry

Name:

Date:

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A student has a 30 cm³ sample of dilute hydrochloric acid.
The concentration of the acid is 1 mol dm⁻³.
He wants the acid to have a concentration of 0.1 mol dm⁻³.
How much water does he need to add to the 30 cm³ sample of acid?

A 30 cm³ B 270 cm³ C 300 cm³ D 330 cm³

Correct Answer: A B C D

Explanation:

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Which of the following is correctly balanced?

A. $\text{Fe} + \text{H}_2\text{SO}_4 \rightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{H}_2$
B. $2 \text{Fe} + \text{H}_2\text{SO}_4 \rightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{H}_2$
C. $2 \text{Fe} + 3 \text{H}_2\text{SO}_4 \rightarrow \text{Fe}_2(\text{SO}_4)_3 + 3\text{H}_2$
D. $\text{Fe} + 3 \text{H}_2\text{SO}_4 \rightarrow \text{Fe}_2(\text{SO}_4)_3 + 3\text{H}_2$

Correct Answer: A B C D

Explanation:

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Which symbol equation for the reaction between sodium carbonate (Na₂CO₃) and hydrochloric acid (HCl) correctly represents the word equation below?
Sodium carbonate (solid) + Hydrochloric acid (solution) → Sodium chloride (solution) + Carbon dioxide (gas) + Water (liquid)

A $\text{Na}_2\text{CO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow 2\text{NaCl}(\text{aq}) + 2\text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$ B $\text{Na}_2\text{CO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow 2\text{NaCl}(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
C $\text{Na}_2\text{CO}_3(\text{s}) + \text{HCl}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$ D $2\text{Na}_2\text{CO}_3(\text{s}) + 4\text{HCl}(\text{aq}) \rightarrow 4\text{NaCl}(\text{aq}) + \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$

Correct Answer: A B C D

Explanation:

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Calculate the volume of 121 g carbon dioxide gas at room temperature and 1 atmosphere pressure.

- A. 24 dm³
- B. 2904 dm³
- C. 66 dm³
- D. 8.73 dm³

Correct Answer: A B C D

Explanation:

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Which one of the following samples of gas occupies the largest volume?

- A 1.0 g of ozone (O₃) at 100 kPa and 300 K
- B 1.0 g of oxygen at 100 kPa and 300 K
- C 1.0 g of water vapour at 250 kPa and 450 K
- D 1.0 g of methane at 333 kPa and 500 K

Correct Answer: A B C D

Explanation:

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Calculate the volume of carbon dioxide given off, measured at room temperature and pressure, when 9.0 g glucose ferments.

$$\text{C}_6\text{H}_{12}\text{O}_6(\text{aq}) \rightarrow 2\text{C}_2\text{H}_5\text{OH}(\text{aq}) + 2\text{CO}_2(\text{g})$$

Molar volume at room temperature and pressure = 24000 cm³ mol⁻¹

A_r values: C = 12, H = 1, O = 16.

A
 600 cm³

B
 1200 cm³

C
 2400 cm³

D
 4800 cm³

Correct Answer: A B C D

Explanation:

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The Avogadro constant is $6.0 \times 10^{23} \text{ mol}^{-1}$.
 Therefore the number of **atoms** in 1 mol of carbon dioxide is

☐ A 2.0×10^{23}

☐ B 6.0×10^{23}

☐ C 1.2×10^{24}

☐ D 1.8×10^{24}

Correct Answer: A B C D

Explanation:

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Which of the following contains the most chloride ions? [1 mark]

A 10 cm³ of 3.30×10^{-2} mol dm⁻³ aluminium chloride solution ☐

B 20 cm³ of 5.00×10^{-2} mol dm⁻³ calcium chloride solution ☐

C 30 cm³ of 3.30×10^{-2} mol dm⁻³ hydrochloric acid ☐

D 40 cm³ of 2.50×10^{-2} mol dm⁻³ sodium chloride solution ☐

Correct Answer: A B C D

Explanation:

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The molar mass of Magnesium is 24g.
12g of magnesium contains the same number of particles as:

A: 6 g of Carbon -12

B: 12 g of Carbon -12

C: 24 g of Carbon -12

D: 6.02×10^{23} g of Carbon -12

Correct Answer: A B C D

Explanation:

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5 The relative atomic mass of chlorine is 35.5.
What is the mass of 2 moles of chlorine gas?

A 17.75g B 35.5g C 71g D 142g

Correct Answer: A B C D

Explanation:

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In a titration, a student uses sulfuric acid to neutralise sodium hydroxide.

$$\text{H}_2\text{SO}_4(\text{aq}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{Na}_2\text{SO}_4(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$$

He finds that 25.00 cm³ of 0.0880 mol dm⁻³ aqueous sodium hydroxide, NaOH, is neutralised by 17.60 cm³ of dilute sulfuric acid, H₂SO₄.

What is the concentration of the sulfuric acid?

A 0.250 mol dm⁻³ B 0.125 mol dm⁻³

C 0.0625 mol dm⁻³ D 0.00125 mol dm⁻³

Correct Answer: A B C D

Explanation:

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For which solution would it be scientifically correct to say that it contains 0.0050 mol of sodium hydroxide?

A 25.0 cm³ of 0.20 mol dm⁻³ B 10 cm³ of 0.5 mol dm⁻³

C 20.00 cm³ of 0.250 mol dm⁻³ D 2.5 cm³ of 0.50 mol dm⁻³

Correct Answer: A B C D

Explanation:

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What is the percentage yield for this reaction?

Lucianna carried out a reaction and produced 74g of copper sulphate. She had reacted 40g of copper oxide (NaOH) with 49g of sulphuric acid (H₂SO₄). The predicted yield is 80g. What is the percentage yield?

a) 92.50%
b) 50.00%
c) 61.25%
d) 66.20%

Correct Answer: A B C D

Explanation:

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What is the percentage yield for this reaction?

Aidan carried out a reaction and produced 111g of sodium chloride. He had reacted 80g of Sodium hydroxide (NaOH) with 73g of hydrochloric acid (HCl). The predicted yield is 117g. What is the percentage yield?

a) 68.4%
b) 94.9%
c) 62.4%
d) 64.9%

Correct Answer: A B C D

Explanation:

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Sodium hydroxide neutralises sulphuric acid.

The equation for the reaction is:

$$2\text{NaOH} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$$

A student used a pipette to add 25.0 cm³ of sodium hydroxide of unknown concentration to a conical flask.

The student carried out a titration and found that 27.12 cm³ of 0.100 mol / dm³ sulphuric acid was needed to neutralise the sodium hydroxide.

Calculate the concentration of the sodium hydroxide.

A. 0.217 mol / dm³
B. 0.108 mol / dm³
C. 0.00542 mol / dm³
D. 0.0184 mol / dm³

Correct Answer: A B C D

Explanation:

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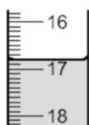
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C5 Titrations

A titration is completed using a burette.

Identify the **correct** reading on the burette.



A 16.9 cm³ **B** 16.8 cm³
C 17.2 cm³ **D** Other reading

Correct Answer: A B C D

Explanation:

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Which is the correct percentage element in the compound?

What is the percentage of oxygen in calcium hydroxide, Ca(OH)₂?

(RAM Ca = 40, O = 16, H = 1)

a) 27.6%
b) 28.1%
c) 54%
d) 43.2%

Correct Answer: A B C D

Explanation:

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What is the empirical formula for a substance made up of 52.2% carbon, 13.0% hydrogen and 34.8% oxygen?

A_r values: C = 12, H = 1, O = 16.

A **B** **C** **D**

C₂H₆O C₃H₆O CH₃O₂ C₄H₁₃O₂

Correct Answer: A B C D

Explanation:

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